

Project Information

For: Michael and Raquel Wachter

Notes: Ref. Man. J per site visit on 4/13/26 and plan 7/1/16 | Assumed 'average' duct seal | R-8 flex insulation | Duct config.: S/A = 'trunk & branch, perimeter outlets' & R/A = 'radial' | R-20 open cell foam @ rafters | Glazing NFRC: N/A - all windows/doors modeled as double pane, low-E, .32-U, .25-SHGC | Orientation per 'Site Plan', verified via site visit | Fresh-air: Not visibly installed, but

Design Information

Weather: Houston Dunn, TX, US

Winter Design Conditions

Outside db	36 °F
Inside db	70 °F
Design TD	34 °F

Ventilation Method

Heating Summary

Structure	32629 Btuh
Ducts (R-8.0)	6452 Btuh
Central vent (84 cfm)	3103 Btuh
Humidification	0 Btuh
Piping	0 Btuh
Equipment load	42183 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	1 (Average)

	Heating	Cooling
Area (ft²)	3940	3940
Volume (ft³)	38678	38678
Air changes/hour	0.40	0.19
Equiv. AVF (cfm)	231	71

Heating Equipment Summary

Make	n/a
Trade	n/a
Model	n/a
AHRI ref.	n/a

Efficiency	n/a
Heating input	
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	n/a

Summer Design Conditions

Outside db	100 °F
Inside db	75 °F
Design TD	25 °F
Daily range	M
Relative humidity	50 %
Moisture difference	39 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	29801 Btuh
Ducts (R-8.0)	4673 Btuh
Central vent (84 cfm)	2284 Btuh
Blower	0 Btuh
Use manufacturer's data	y
Rate/swing multiplier	1.00
Equipment sensible load	36757 Btuh

Latent Cooling Equipment Load Sizing

Structure	3091 Btuh
Ducts	2506 Btuh
Central vent (84 cfm)	2246 Btuh
Equipment latent load	5597 Btuh
Equipment Total Load (Sen+Lat)	42354 Btuh
Req. total capacity at 0.70 SHR	4.4 ton

Cooling Equipment Summary

Make	n/a
Trade	n/a
Cond	n/a
Coil	n/a
AHRI ref.	n/a
Efficiency	n/a
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.

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Design Information

Weather: Houston Dunn, TX, US

Winter Design Conditions

Outside db	36 °F
Inside db	70 °F
Design TD	34 °F

Ventilation Method ASHRAE 62.2-2010

Heating Summary

Structure	13933 Btuh
Ducts (R-8.0)	2695 Btuh
Central vent (24 cfm)	892 Btuh
Outside air	
Humidification	0 Btuh
Piping	0 Btuh
Equipment load	17521 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	1 (Average)

	Heating	Cooling
Area (ft²)	1677	1677
Volume (ft³)	16764	16764
Air changes/hour	0.36	0.11
Equiv. AVF (cfm)	99	30

Heating Equipment Summary

Make	Carrier
Trade	CARRIER
Model	59*P5A080E17**16
AHRI ref	

Efficiency	96 AFUE
Heating input	80000 Btuh
Heating output	76800 Btuh
Temperature rise	103 °F
Actual air flow	683 cfm
Air flow factor	0.041 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

Summer Design Conditions

Outside db	100 °F
Inside db	75 °F
Design TD	25 °F
Daily range	M
Relative humidity	50 %
Moisture difference	39 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	13199 Btuh
Ducts (R-8.0)	1948 Btuh
Central vent (24 cfm)	657 Btuh
Outside air	
Blower	0 Btuh
Use manufacturer's data	y
Rate/swing multiplier	1.00
Equipment sensible load	15803 Btuh

Latent Cooling Equipment Load Sizing

Structure	1011 Btuh
Ducts	1050 Btuh
Central vent (24 cfm)	646 Btuh
Outside air	
Equipment latent load	2706 Btuh

Equipment Total Load (Sen+Lat)	18510 Btuh
Req. total capacity at 0.70 SHR	1.9 ton

Cooling Equipment Summary

Make	Carrier
Trade	CARRIER
Cond	24ABC630A*030*
Coil	CNPH*3117AL*
AHRI ref	9169630
Efficiency	15.5 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	683 cfm
Air flow factor	0.045 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.85

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Design Information

Weather: Houston Dunn, TX, US

Winter Design Conditions

Outside db	36 °F
Inside db	70 °F
Design TD	34 °F

Ventilation Method ASHRAE 62.2-2010

Heating Summary

Structure	18695 Btuh
Ducts (R-8.0)	3756 Btuh
Central vent (60 cfm)	2211 Btuh
Outside air	
Humidification	0 Btuh
Piping	0 Btuh
Equipment load	24662 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	1 (Average)

	Heating	Cooling
Area (ft²)	2262	2262
Volume (ft³)	21914	21914
Air changes/hour	0.36	0.11
Equiv. AVF (cfm)	132	41

Heating Equipment Summary

Make	Carrier
Trade	CARRIER
Model	59*P5A080E17**16
AHRI ref	

Efficiency	96 AFUE
Heating input	80000 Btuh
Heating output	76800 Btuh
Temperature rise	81 °F
Actual air flow	869 cfm
Air flow factor	0.039 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

Summer Design Conditions

Outside db	100 °F
Inside db	75 °F
Design TD	25 °F
Daily range	M
Relative humidity	50 %
Moisture difference	39 gr/lb

Sensible Cooling Equipment Load Sizing

Structure	16602 Btuh
Ducts (R-8.0)	2725 Btuh
Central vent (60 cfm)	1627 Btuh
Outside air	
Blower	0 Btuh
Use manufacturer's data	y
Rate/swing multiplier	1.00
Equipment sensible load	20954 Btuh

Latent Cooling Equipment Load Sizing

Structure	2080 Btuh
Ducts	1456 Btuh
Central vent (60 cfm)	1600 Btuh
Outside air	
Equipment latent load	5137 Btuh

Equipment Total Load (Sen+Lat)	26091 Btuh
Req. total capacity at 0.70 SHR	2.5 ton

Cooling Equipment Summary

Make	Carrier
Trade	CARRIER
Cond	24ABC636A*031*
Coil	CNPH*4321AL*
AHRI ref	203474660
Efficiency	15.5 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	869 cfm
Air flow factor	0.045 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.80

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Project Information

For: Michael and Raquel Wachter

Design Information

	Htg	Clg	Infiltration	
Outside db (°F)	36	100	Method	Simplified
Inside db (°F)	70	75	Construction quality	Average
Design TD (°F)	34	25	Fireplaces	1 (Average)
Daily range	-	M		
Inside humidity (%)	30	50		
Moisture difference (gr/lb)	8	39		

HEATING EQUIPMENT

Make	n/a
Trade	n/a
Model	n/a
AHRI ref	n/a
Efficiency	n/a
Heating input	
Heating output	0 Btuh
Temperature rise	0 °F
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	n/a

COOLING EQUIPMENT

Make	n/a
Trade	n/a
Cond	n/a
Coil	n/a
AHRI ref	n/a
Efficiency	n/a
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	0 cfm
Air flow factor	0 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
AHU1	1677	16628	15147	683	683
AHU2	2262	22452	19327	869	869
Entire House	3940	39080	34474	1552	1552
Other equip loads		3103	2284		
Equip. @ 1.00 RSM			36757		
Latent cooling			5597		
TOTALS	3940	42183	42354	1552	1552

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Design Information

	Htg	Clg	Infiltration	
Outside db (°F)	36	100	Method	Simplified
Inside db (°F)	70	75	Construction quality	Average
Design TD (°F)	34	25	Fireplaces	1 (Average)
Daily range	-	M		
Inside humidity (%)	30	50		
Moisture difference (gr/lb)	8	39		

HEATING EQUIPMENT

Make	Carrier
Trade	CARRIER
Model	59*P5A080E17**16
AHRI ref	
Efficiency	96 AFUE
Heating input	80000 Btuh
Heating output	76800 Btuh
Temperature rise	103 °F
Actual air flow	683 cfm
Air flow factor	0.041 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

COOLING EQUIPMENT

Make	Carrier
Trade	CARRIER
Cond	24ABC630A*030*
Coil	CNPH*3117AL*
AHRI ref	9169630
Efficiency	15.5 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	683 cfm
Air flow factor	0.045 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.85

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
BREAKFAST	143	2211	1425	91	64
KITCHEN	243	1710	2913	70	131
FAMILY	574	5129	4126	211	186
STUDY	161	1965	3152	81	142
BATH 4	62	474	299	19	13
CLOS.	9	17	11	1	1
FOYER	180	1765	1355	72	61
DINING	236	2844	1592	117	72
B.P.	39	456	234	19	11
PANTRY	30	57	39	2	2

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.

AHU1	1677	16628	15147	683	683
Other equip loads		892	657		
Equip. @ 1.00 RSM			15803		
Latent cooling			2706		
TOTALS	1677	17521	18510	683	683

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.



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...4308 Vivian Street\Load\4308 Vivian Street.rup Calc = MJ8 Front Door faces: S

Project Information

For: Michael and Raquel Wachter

Design Information

	Htg	Clg	Infiltration	
Outside db (°F)	36	100	Method	Simplified
Inside db (°F)	70	75	Construction quality	Average
Design TD (°F)	34	25	Fireplaces	1 (Average)
Daily range	-	M		
Inside humidity (%)	30	50		
Moisture difference (gr/lb)	8	39		

HEATING EQUIPMENT

Make	Carrier
Trade	CARRIER
Model	59*P5A080E17**16
AHRI ref	
Efficiency	96 AFUE
Heating input	80000 Btuh
Heating output	76800 Btuh
Temperature rise	81 °F
Actual air flow	869 cfm
Air flow factor	0.039 cfm/Btuh
Static pressure	0 in H2O
Space thermostat	

COOLING EQUIPMENT

Make	Carrier
Trade	CARRIER
Cond	24ABC636A*031*
Coil	CNPH*4321AL*
AHRI ref	203474660
Efficiency	15.5 SEER
Sensible cooling	0 Btuh
Latent cooling	0 Btuh
Total cooling	0 Btuh
Actual air flow	869 cfm
Air flow factor	0.045 cfm/Btuh
Static pressure	0 in H2O
Load sensible heat ratio	0.80

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
M. BED	338	3883	2897	150	130
M. BATH	279	2100	1285	81	58
M. W/C	30	131	92	5	4
SECRET CLOS.	86	1298	657	50	30
M. WIC	199	1877	2076	73	93
BED 4	188	1705	2251	66	101
B4 WIC	29	195	130	8	6
DRESS. 3	58	828	424	32	19
BATH 3	39	901	433	35	19
BED 3	267	3175	2333	123	105
BED 2	195	1600	1462	62	66
B2 WIC	33	833	396	32	18
BATH 2	57	694	345	27	15
GAME	260	832	1365	32	61

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.

GAME SLP	34	1210	1881	47	85
UTIL	80	694	951	27	43
UPPER HALL	48	218	148	8	7
STAIRS UP	42	275	201	11	9
AHU2	2262	22452	19327	869	869
Other equip loads		2211	1627		
Equip. @ 1.00 RSM			20954		
Latent cooling			5137		
TOTALS	2262	24662	26091	869	869

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Project Information

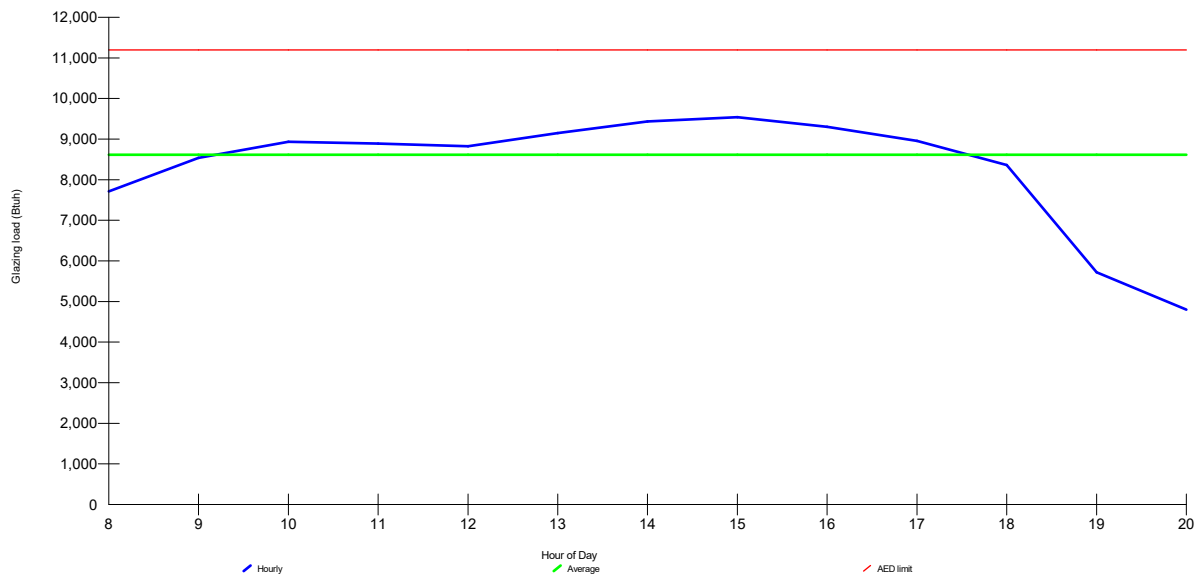
For: Michael and Raquel Wachter

Design Conditions

Location:			Indoor:	Heating	Cooling
Houston Dunn, TX, US			Indoor temperature (°F)	70	75
Elevation: 227 ft			Design TD (°F)	34	25
Latitude: 30°N			Relative humidity (%)	30	50
Outdoor:	Heating	Cooling	Moisture difference (gr/lb)	7.6	39.5
Dry bulb (°F)	36	100	Infiltration:		
Daily range (°F)	-	18 (M)			
Wet bulb (°F)	-	77			
Wind speed (mph)	15.0	7.5			

Test for Adequate Exposure Diversity

Hourly Glazing Load



Maximum hourly glazing load exceeds average by 10.8%.

House has adequate exposure diversity (AED), based on AED limit of 30%.

AED excursion: 0 Btu/h

Project Information

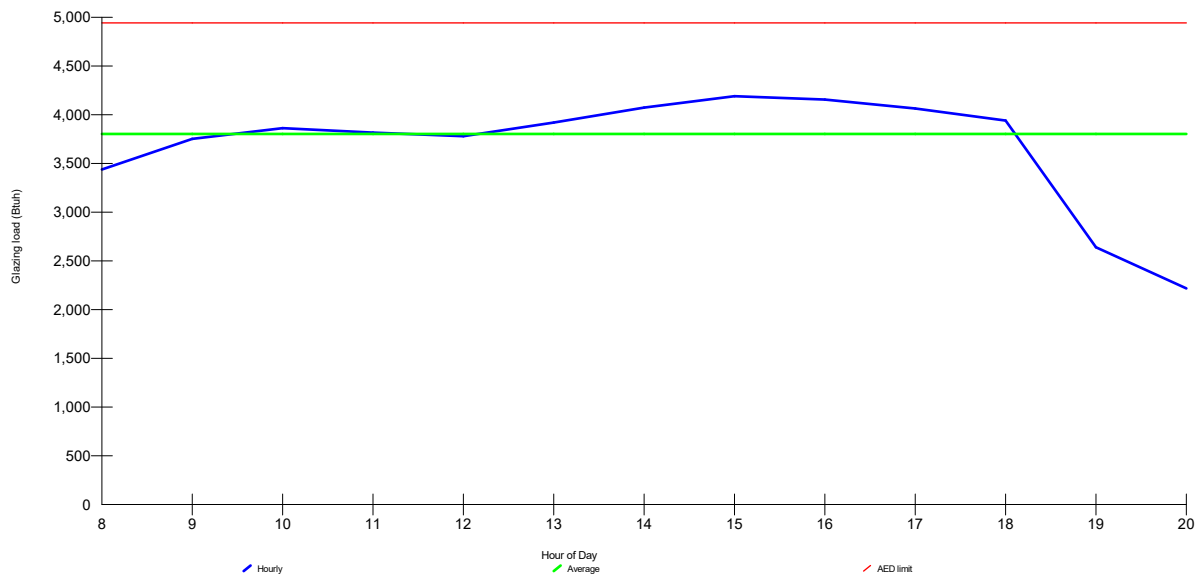
For: Michael and Raquel Wachter

Design Conditions

Location:		Indoor:		Heating	Cooling
Houston Dunn, TX, US		Indoor temperature (°F)		70	75
Elevation: 227 ft		Design TD (°F)		34	25
Latitude: 30°N		Relative humidity (%)		30	50
Outdoor:		Infiltration:		7.6	39.5
Heating	Cooling				
Dry bulb (°F)	36	Moisture difference (gr/lb)			
Daily range (°F)	-				
Wet bulb (°F)	-				
Wind speed (mph)	15.0				

Test for Adequate Exposure Diversity

Hourly Glazing Load



Maximum hourly glazing load exceeds average by 10.2%.

Zone has adequate exposure diversity (AED), based on AED limit of 30%.

AED excursion: 0 Btu/h

Project Information

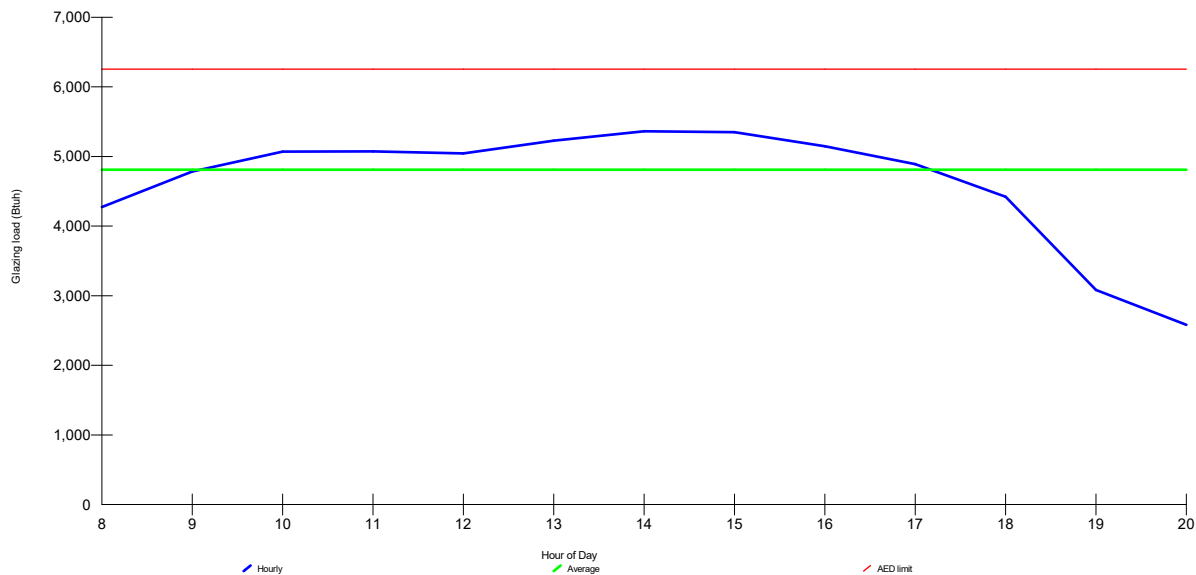
For: Michael and Raquel Wachter

Design Conditions

Location:		Indoor:		Heating	Cooling
Houston Dunn, TX, US		Indoor temperature (°F)		70	75
Elevation: 227 ft		Design TD (°F)		34	25
Latitude: 30°N		Relative humidity (%)		30	50
Outdoor:		Infiltration:		7.6	39.5
Heating	Cooling				
Dry bulb (°F)	36	Moisture difference (gr/lb)			
Daily range (°F)	-				
Wet bulb (°F)	-				
Wind speed (mph)	15.0				

Test for Adequate Exposure Diversity

Hourly Glazing Load



Maximum hourly glazing load exceeds average by 11.4%.

Zone has adequate exposure diversity (AED), based on AED limit of 30%.

AED excursion: 0 Btuh

Project Information

For: Michael and Raquel Wachter

Design Conditions

Location:

Houston Dunn, TX, US
Elevation: 227 ft
Latitude: 30°N

Outdoor:

Dry bulb (°F)
Daily range (°F)
Wet bulb (°F)
Wind speed (mph)

Heating

36
-
-
15.0

Cooling

100
18 (M)
77
7.5

Indoor:

Indoor temperature (°F)
Design TD (°F)
Relative humidity (%)
Moisture difference (gr/lb)

Heating

70
34
30
7.6

Cooling

75
25
50
39.5

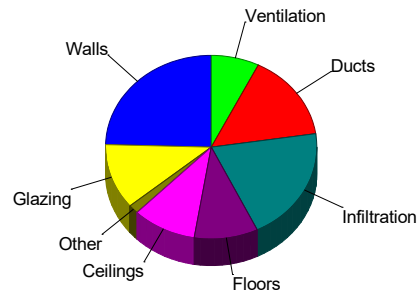
Infiltration:

Method
Construction quality
Fireplaces

Simplified
Average
1 (Average)

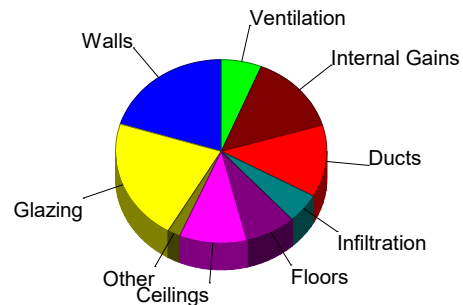
Heating

Component	Btuh/ft²	Btuh	% of load
Walls	2.9	10354	24.5
Glazing	10.8	4855	11.5
Doors	13.1	648	1.5
Ceilings	1.8	4150	9.8
Floors	1.8	4112	9.7
Infiltration	2.5	8510	20.2
Ducts		6452	15.3
Piping		0	0
Humidification		0	0
Ventilation		3103	7.4
Adjustments		0	0
Total		42183	100.0



Cooling

Component	Btuh/ft²	Btuh	% of load
Walls	2.1	7413	20.2
Glazing	17.5	7892	21.5
Doors	14.5	715	1.9
Ceilings	1.6	3798	10.3
Floors	1.2	2859	7.8
Infiltration	0.6	1922	5.2
Ducts		4673	12.7
Ventilation		2284	6.2
Internal gains		5201	14.1
Blower		0	0
Adjustments		0	0
Total		36757	100.0



Latent Cooling Load = 5597 Btuh
Overall U-value = 0.183 Btuh/ft²-°F, Window / Floor Area = 11.4 %

Data entries checked.

Project Information

For: Michael and Raquel Wachter

Design Conditions

Location:

Houston Dunn, TX, US
Elevation: 227 ft
Latitude: 30°N

Outdoor:

Dry bulb (°F)
Daily range (°F)
Wet bulb (°F)
Wind speed (mph)

Heating

36
-
-
15.0

Cooling

100
18 (M)
77
7.5

Indoor:

Indoor temperature (°F)
Design TD (°F)
Relative humidity (%)
Moisture difference (gr/lb)

Heating

70
34
30
7.6

Cooling

75
25
50
39.5

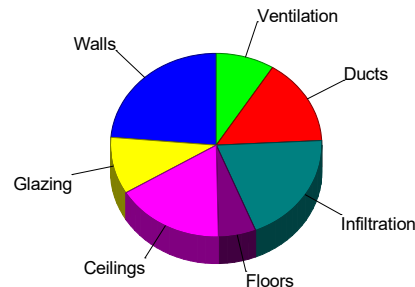
Infiltration:

Method
Construction quality
Fireplaces

Simplified
Average
1 (Average)

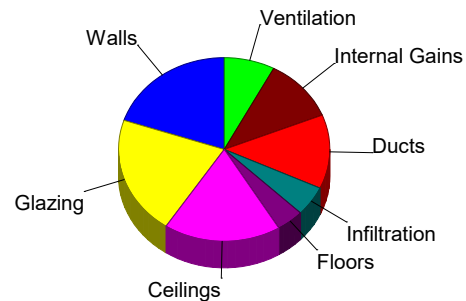
Heating

Component	Btuh/ft²	Btuh	% of load
Walls	2.9	5829	23.6
Glazing	10.8	2480	10.1
Doors	0	0	0
Ceilings	1.8	4102	16.6
Floors	2.2	1423	5.8
Infiltration	2.5	4862	19.7
Ducts		3756	15.2
Piping		0	0
Humidification		0	0
Ventilation		2211	9.0
Adjustments		0	0
Total		24662	100.0



Cooling

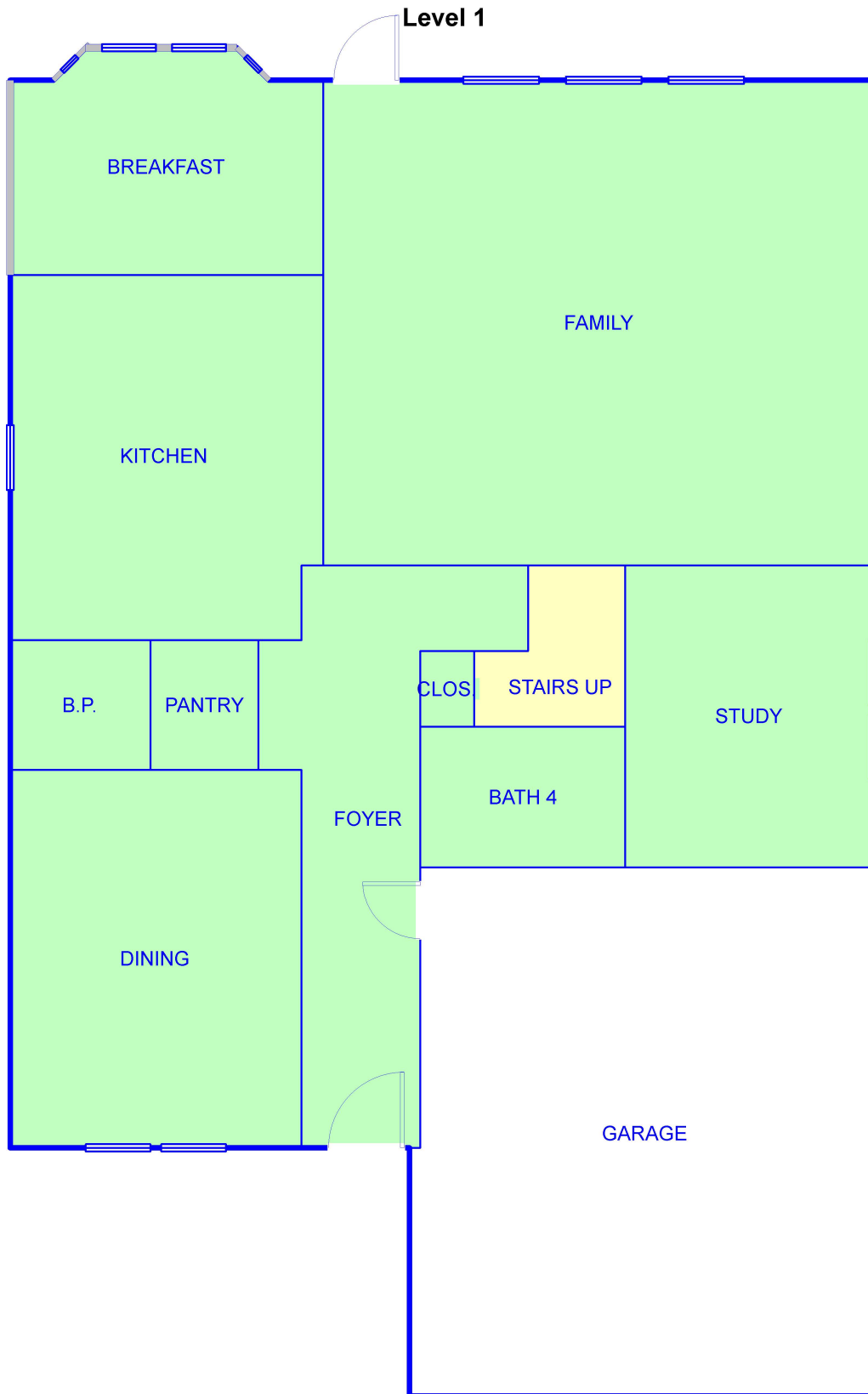
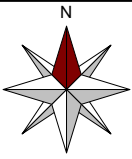
Component	Btuh/ft²	Btuh	% of load
Walls	2.1	4160	19.9
Glazing	19.0	4375	20.9
Doors	0	0	0
Ceilings	1.6	3755	17.9
Floors	1.4	880	4.2
Infiltration	0.6	1098	5.2
Ducts		2725	13.0
Ventilation		1627	7.8
Internal gains		2333	11.1
Blower		0	0
Adjustments		0	0
Total		20954	100.0



Latent Cooling Load = 5137 Btuh

Overall U-value = 0.241 Btuh/ft²-°F, Window / Floor Area = 10.2 %

Data entries checked.



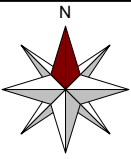
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Performed by TPD-3 for:
Michael and Raquel Wachter

Texas Performance Designs, LLC.

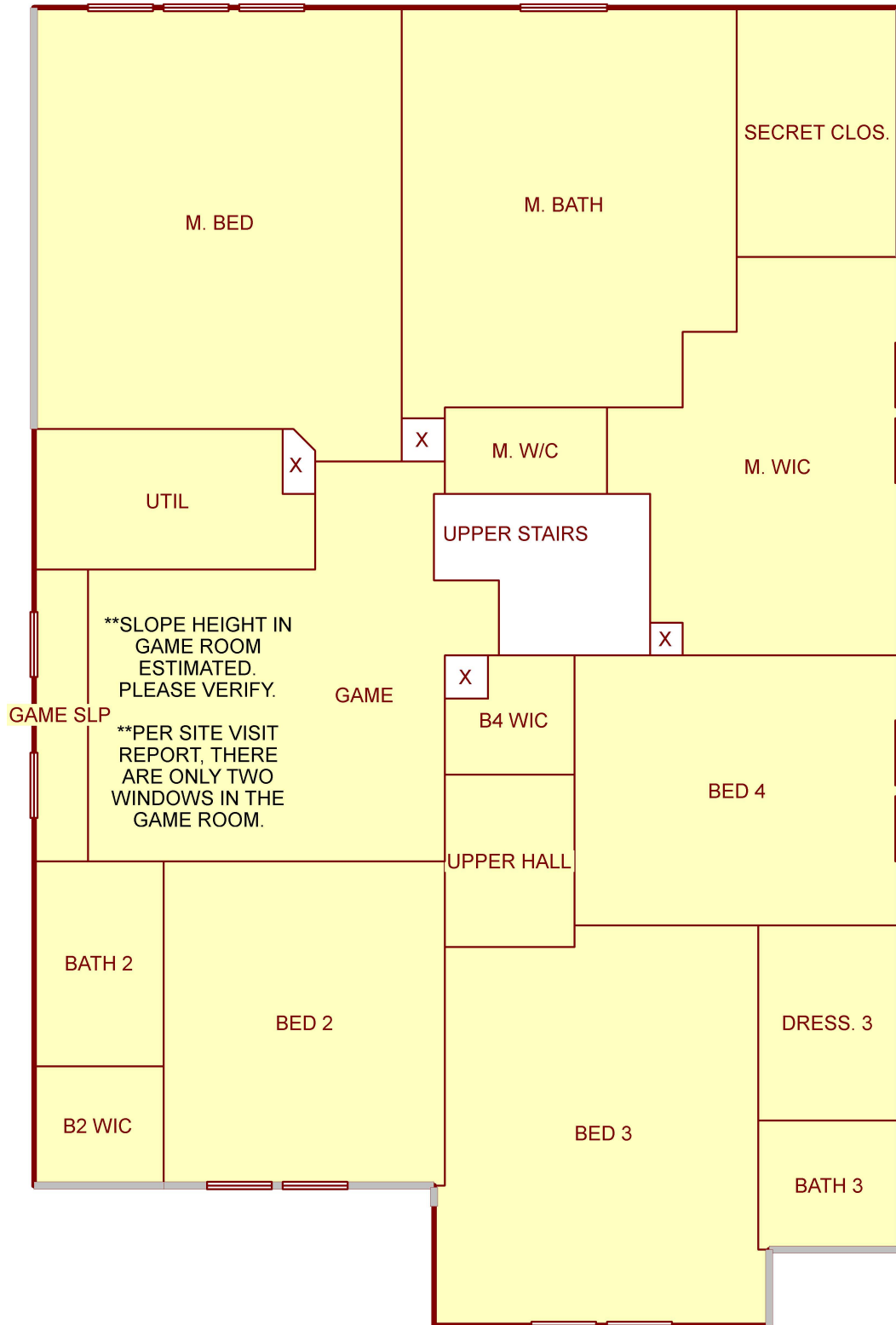
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Josh@TxPerformanceDesigns.com

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Level 2



Job #: 4308 Vivian Street
Performed by TPD-3 for:
Michael and Raquel Wachter

Texas Performance Designs, LLC.

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Josh@TxPerformanceDesigns.com

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